



**UG Program in Electronics and Communication
(Advanced Communication Technology)**

Experiment No. 3

Linear block code

3.1 Aim: Write python program for Error detection and correction using Linear block code generation and error detection.

3.2 Course Outcome: Examine various error detection and correction codes and error control.

3.3 Learning Objectives: Students will be able

- Understand the Fundamentals of Linear Block Codes

3.4 Related Theory:

Noise or Error is the main problem in the signal, which disturbs the reliability of the communication system. Error control coding is the coding procedure done to control the occurrences of errors. These techniques help in Error Detection and Error Correction. There are many different error correcting codes depending upon the mathematical principles applied to them. But historically, these codes have been classified into Linear block codes and Convolution codes.

Linear Block Codes

In the linear block codes, the parity bits and message bits have a linear combination, which means that the resultant code word is the linear combination of any two code words.

Let us consider some blocks of data, which contains k bits in each block. These bits are mapped with the blocks which has n bits in each block. Here n is greater than k . The transmitter adds redundant bits which are $n-k$ bits. The ratio k/n is the code rate. It is denoted by r and the value of r is $r < 1$.



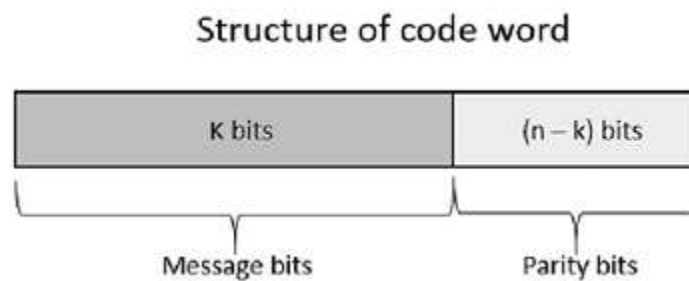
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The $n-k$ bits added here, are parity bits. Parity bits help in error detection and error correction, and also in locating the data. In the data being transmitted, the left most bits of the code word correspond to the message bits, and the right most bits of the code word correspond to the parity bits.

Systematic Code

Any linear block code can be a systematic code, until it is altered. Hence, an unaltered block code is called as a systematic code.

Following is the representation of the structure of code word, according to their allocation.



If the message is not altered, then it is called as systematic code. It means, the encryption of the data should not change the data.

3.5 Program and Output:

Program:

```
import numpy as np
# Generator matrix G
G = np.array([
```

```
[1, 0, 0, 1, 1, 1],
```

```
[0, 1, 0, 1, 1, 0],
```



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```
[0, 0, 1, 0, 1, 1]

])
# Message
message = np.array([0, 1, 1]) # 011
# Encode: c = m * G (mod 2)
codeword = np.dot(message, G) % 2
print("Message:", message)

print("Codeword:", codeword)
# Extract P from G
P = G[:, 3:]
# Identity matrix
I = np.eye(3, dtype=int)
# Parity check matrix H
H = np.hstack((P.T, I))
print("Parity Check Matrix H:\n", H)
# Received sequence
received = np.array([1, 0, 1, 1, 0, 1]) # 101101
# Calculate syndrome: s = r * H^T (mod 2)
syndrome = np.dot(received, H.T) % 2
print("Received:", received)
print("Syndrome:", syndrome)
# Find error position
error_position = -1
for i in range(H.shape[1]):

    if np.array_equal(H[:, i], syndrome):

        error_position = i

        break
# Correct error
if error_position != -1:

    print("Error at position:", error_position)

    received[error_position] ^= 1 # flip bit
print("Corrected codeword:", received)
```

Output:



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Message: [0 1 1]

Codeword: [0 1 1 1 0 1]

Parity Check Matrix H:

[[1 1 0 1 0 0]

[1 1 1 0 1 0]

[1 0 1 0 0 1]]

Received: [1 0 1 1 0 1]

Syndrome: [0 0 1]

Error at position: 5

Corrected codeword: [1 0 1 1 0 0]

3.6 Conclusion:

This program demonstrates encoding a message using a generator matrix and understanding the fundamentals of linear block codes. It detects transmission errors by computing the syndrome using the parity-check matrix. If an error is found, it locates and corrects the bit, ensuring reliable data transmission.

3.7 Questions:



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1) Write a short note on Systematic and Non-systematic codes.

→ Write a short note on Systematic and Non-Systematic codes.

→ Systematic codes:

In a Systematic code, the original message bits appear explicitly and unchanged within the codeword. The codeword is simply the message bits followed by the redundant parity bits. This means the structure of the codeword can be clearly divided into two distinct parts.

- Message part - the original k information bits
- parity part - the $n-k$ redundant bits appended for error control.

Non-Systematic codes:

In a non-systematic code, the original message bits do not appear explicitly in the codeword. Instead, all n bits of the codeword are a transformed version of the message & no part of the codeword directly corresponds to the original message bits.

Advantages:

- Can sometimes offer better distance properties for certain code constructions.
- Used in specific applications like Turbo codes & certain convolutional codes.